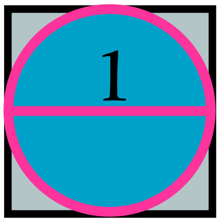
Round

If you're a whizz at maths, try this tricky puzzle: which is bigger in area – the blue ring on these pages or the three central rings (red, pink, and white combined)? Hint: the area of a circle is πr^2)

A slice of pi

The Greeks were also fascinated by pi – the ratio between the circumference (distance around) and the diameter (distance across) of a circle. Pi is impossible to measure exactly, but the Greeks had a go anyway, by comparing circles to other shapes …

… here's a circle inside a square:

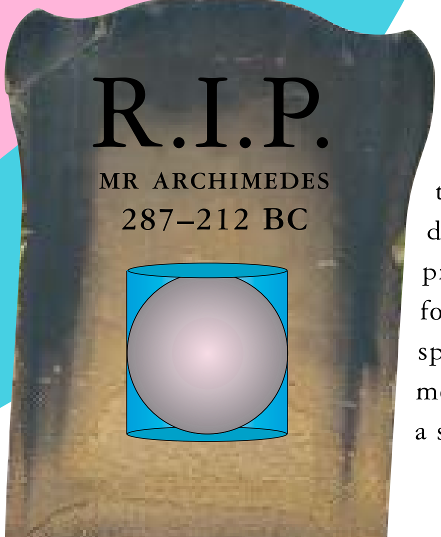


If the diameter is 1, the circumference must be pi. The distance around the square must be 4, since each side is 1. So pi is less than 4.

Here's a circle outside a hexagon:

The diameter is still 1 and the circumference is still pi. What about the distance around the hexagon? The radius of the circle is 0.5, so each side of the hexagon must be 0.5 long too. That means the distance around the hexagon is 3. The circle is a bit bigger than the hexagon, so pi must be a bit more than 3.

The Greek mathematician Archimedes carried on like this, getting closer to pi as he moved from hexagon to octagon and so on. He ended up drawing shapes with 96 sides and so proved that pi is between $^{223}/_{71}$ and $^{220}/_{70}$. And then he gave up.



Death by maths

According to legend, Archimedes was killed by a Roman soldier who lost his temper when Archimedes refused to stop drawing circles in the ground. Archimedes' proudest achievement was finding the formula for the volume and surface area of a sphere, which he found by studying wooden models of them. To honour this, he asked for a sphere and cylinder to be carved on his tomb.

A round world

Another clever Greek mathematician was Eratosthenes, who lived in Egypt around 1250 BC. He used circular maths to prove Earth is round and even measured its size, which was amazing for a time when most people thought Earth was flat! So how did he do it?

Eratosthenes heard that at Syene (now Aswan) in southern Egypt, sunlight shone straight down wells in midsummer, when the Sun was directly overhead. So, on the same day, he measured the angle of the Sun in Alexandria in the north, and found it hit the ground at 7.2°, casting a small shadow. Eratosthenes realized this was because Earth was curved. As the Sun's rays are parallel, he also realized that two lines drawn straight to the centre of the Earth from these two places would meet at an angle of 7.2°. This is exactly a fiftieth of a circle (360°). So he just multiplied the distance between the two places – 800 km (500 miles) – by 50 to get 40,000 km (25,000 miles) for the distance around Earth. It remained the most accurate estimate for 2000 years.